NMTC Problem-Set 3

Instructions: For this problem-set, it is assumed that you know the following topics to some degree.

1. Angle Chasing
2. Basic Combinatorics
3. Basic Divisibility, Euler's Generalization of Fermat's Little Theorem
4. Proof by Contradiction
5. Basic methods of solving Functional Equations, including induction, injectivity/surjectivity, etc.

This problem-set is meant to be of the same level as the NMTC descriptive questions and above that. Problems marked at the end with (*) are a bit more tough (and fun!), while problems marked with (**) are tougher, and so on.
These problems are meant to build your problem-solving skills, and as such, they are open-book. You are encouraged to use any textbook, video, or other online or offline resource as long as it doesn't give away the complete solution.
The sources to each applicable problem will be revealed with the solutions.

Problem 1: Some Functional Equations

1. Find all functions $f : \mathbb{N} \to \mathbb{N}$, for which $f(n)$ is the square of an integer for all $n \in \mathbb{N}$, and that satisfy:

$$f(m+n) = f(m) + f(n) + 2mn$$

2. Find all functions $f : \mathbb{N} \to \mathbb{N}$, such that:

$$f(f(m) + f(n)) = m + n$$

for every two natural numbers m and n.

Problem 2: Combinatorics

1. Several real numbers are written on the board. At each step, we are allowed to choose two numbers a, b on the board, such that $a - b \geq 2$, and replace them with $a - 1, b + 1$. Prove that we can do only finitely many steps.

2. In a cursed forest, there are 55 wolves, 17 cats, and 6 sparrows. Every time a wolf eats a cat, it becomes a sparrow; every time a wolf eats a sparrow, it becomes a cat; every time a cat eats a sparrow, it becomes a wolf.

   Finally, at some moment, two species are extinct and animals of only one species remain. When this happens, at most how many animals can there be in the forest?

Problem 3: Cyclic Quads!

Let ABC be an isosceles triangle with $AB = AC$. Take a point O inside the triangle (not on the boundary) and let ω be the circle with center O passing through C. Let ω intersect the line BC at D ($\neq C$) and intersect the line AC at E ($\neq C$). Let Γ be the circumcircle of $\triangle AEO$, and suppose Γ intersects ω at F ($\neq E$).

(a) Prove that $AF = AB = AC$.

(b) Suppose that the line ED and the angle bisector of $\angle BAF$ intersect at X. Show that X, F, O, A form a cyclic quadrilateral.

(c) Show that X is the circumcentre of $\triangle BDF$.

Problem 4: Integer Polynomials

Let $P(n)$ be a polynomial with integer coefficients.

(a) Prove that for any integers a, b, we have $a - b \mid P(a) - P(b)$.

(b) Prove that for any integer k and odd integer m, there are infinitely many integers x such that $m \mid P(2^{kx}) - P(2^k)$.

(c) Given that that for any integer k, $k \mid P(2^k)$, find all polynomials $P(n)$.(*)